Inhomogeneous self-consistent mean-field theory of zinc doping of the cuprates

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Contents

1 Introduction 1

2 Topology 2

3 Review of defects in correlated metals 2

4 Reproducing results 2

5 Self-consistent mean-field solution 3

6 Free energy 4

7 Green's function 4

1 Introduction

J.C. Seamus Davis discussed with me the problem of zinc doping in the cuprates. It unknown why the superconductivity is killed by small doping. Seamus suggested I try modelling the situation by switching off the Hubbard U on the zinc site. Copper peroxide (CuO_2) is stabilised by surrounding elements in the cuprate, thought to be irrelevant to superconductivity due to the lack of supercurrent along the third axis. In addition to the stabilisers, the ingredients for a cuprate superconductivity are copper (Cu) whose d-wave orbital provides unconventional hopping, and oxygen (O_2) which staggers the hopping. The minimal model of superconductivity is onsite singlet pairing, and hence the d-wave pairing is irrelevant to the interaction term. More unconventional models of the superconductor may include multiorbital triplet pairing, in which the d-wave and p-wave symmetries would become relevant to the interaction (e.g. nearest-neighbour pairing in the bulk via Monte-Carlo [Kobayashi and Yokoyama REF)]. We consider the zinc-doped minimal model.

H. Alloul *et al* review Defects in correlated metals and superconductors [CITE:arXiv:0711.0877v1 (2018)].

2 Topology

Rosenberg and Manousakis [CITE: Rosenberg, Aryal PRB 100 104522 (2019)] showed topological modes when the interdimer hopping is different from the intradimer hopping, with equal chemical potential.

Have they checked with a rigid shift?

Diagonal hoppings within the (Cu-O)$_4$ cluster

Any nonzero CDW-like density in the Fock channel would give rise to superconductivity.

3 Review of defects in correlated metals

H. Alloul *et al* [CITE:arXiv:0711.0877v1 (2018)] review theories of impurities in d-wave superconductor theories of impurities

4 Reproducing results

The normal state Hamiltonian describing the copper peroxide plane with zinc impurity doping is given by

$$\begin{aligned}\hat{\mathcal{H}}_0 \equiv & \sum_{-N/2\leq x,y\leq N/2}\sum_{\sigma\in\{\uparrow,\downarrow\}}\sum_{m\in\{d,p_x,p_y\}}\left[-\mu_m \hat{c}^{\sigma,m\dagger}_{x,y}\hat{c}^{\sigma,m}_{x,y}\right] + \sum_{-L/2\leq x,y\leq L/2}\sum_{\sigma\in\{\uparrow,\downarrow\}}\left[+V\hat{c}^{\sigma,d\dagger}_{x,y}\hat{c}^{\sigma,d}_{x,y}\right] \\ & + \sum_{-N/2\leq x,y\leq N/2}\sum_{\sigma\in\{\uparrow,\downarrow\}}\left[+t\hat{c}^{\sigma,d\dagger}_{x,y}\hat{c}^{\sigma,p_x}_{x,y} - t\hat{c}^{\sigma,d\dagger}_{x,y}\hat{c}^{\sigma,p_y}_{x,y} - t\hat{c}^{\sigma,p_x\dagger}_{x,y}\hat{c}^{\sigma,d}_{x+1,y} + t\hat{c}^{\sigma,p_y\dagger}_{x,y}\hat{c}^{\sigma,d}_{x,y+1}\right] \quad (1)\end{aligned}$$

where the fermion operator $\hat{c}^{\sigma,m\dagger}_{x,y}$ creates an operator on atomic orbital $m \in \{d_{x^2-y^2}, p_x, p_y\}$ within the unit cell located at $\mathbf{R}_{x,y} = x\mathbf{b}_0\hat{x}+y\mathbf{b}_1\hat{y}$, where $\mathbf{b}_0, \mathbf{b}_1$ are the lattice basis vectors, $\hat{n}^{(\alpha)\dagger}_{x,y}$ is the number operator, $\mu_d \equiv \mu$ and $\mu_{p_x} \equiv \mu_{p_y} \equiv \mu + s$ are the chemical potentials of the copper and oxygen orbitals and s is a rigid shift of the atomic potentials. The parameter V sets the coupling strength of a zinc impurity at the centre of our lattice, and whose size goes from a single site $L = 0$ to a cluster $L > 0$. In the minimal theory, we assume half-filling $\mu = s = 0$ and zero coupling $V = 0$. The interorbital, intra- and inter-unit-cell hoppings strengths are given by $t = 1$, which normalises the model paramters. The boundary conditions are taken to be periodic. The lattice is depicted with $N = 3$ unit cells in Figure 1.

The interacting part of the minimal theory describes a singlet electron-electron interaction on the copper sites, but not on the zinc-doped sites

$$\hat{\mathcal{V}} \equiv \sum_{-N/2 \leq x,y \leq N/2} \sum_{m \in \{d,p_x,p_y\}} \left[+U^m_{x,y} \hat{c}^{\uparrow,m\dagger}_{x,y} \hat{c}^{\downarrow,m\dagger}_{x,y} \hat{c}^{\uparrow,m}_{x,y} \hat{c}^{\downarrow,m}_{x,y} \right], \tag{2}$$

where $U^m_{x,y}$ is the strength of the Hubbard interaction on orbital m, which, in the simplest theory, we let $U^m_{x,y} := U$ for $L/2 < x, y \leq N/2$ and $U^m_{x,y} := 0$ for $0 \leq x, y \leq L/2$. . $U > 0$ describes Coulomb repulsion and $U < 0$ is the attractive BCS (singlet) scenario. The minimal model describing Zn-doped CuO_2 superconducting planes with BCS-like interactions on the Cu sites is then

$$\hat{\mathcal{H}} \equiv \hat{\mathcal{H}}_0 + \hat{\mathcal{V}} \tag{3}$$

We solve this Hamiltonian with inhomogeneous lattice self-consistent mean-field theory.

5 Self-consistent mean-field solution

The Bogoliubov-de Gennes (BdG) mean-field Hamiltonian is

$$\begin{aligned} \hat{\mathcal{V}}_{\rm mf} = & \sum_{\sigma \in \{\uparrow,\downarrow\}} \sum_{m \in \{d,p_x,p_y\}} \sum_{-N/2 \leq x,y \leq N/2} \left[\phi^{\sigma,m}_{x,y} \hat{n}^{\sigma,m}_{x,y} \right] \\ & + \sum_{m \in \{d,p_x,p_y\}} \sum_{-N/2 \leq x,y \leq N/2} \left[\Phi^m_{x,y} \hat{c}^{\uparrow,m}_{x,y} \hat{c}^{\downarrow,m}_{x,y} + \Delta^m_{x,y} \hat{c}^{\uparrow,m\dagger}_{x,y} \hat{c}^{\downarrow,m\dagger}_{x,y} + \Delta^{m*}_{x,y} \hat{c}^{\uparrow,m}_{x,y} \hat{c}^{\downarrow,m}_{x,y} \right], \end{aligned} \tag{4}$$

where ϕ is the *Hartree* term which renormalises the chemical potential, allowing for a Peirl's instability. Φ is a *Fock* term, describing spin-flips on the d-orbitals. The additional *Gorkov* terms Δ_v capture fermion-fermion pairing, relevant to theories of magnetism ($U > 0$) and superconductivity ($U < 0$). These terms, solved self-consistently, are known as inhomogenous *(superconducting) density functional theory, SC-DFT*. The self-consistent solution to the mean field is the saddle point $\langle \hat{\mathcal{V}} \rangle_{\rm mf} = \langle \hat{\mathcal{V}}_{\rm mf} \rangle_{\rm mf}$, in which the mean-field eigenstates $\{E_n, |n\rangle\}$ approximate the state of the interacting theory by minimising (or, unstably, maximising) the free energy, where $E_n \geq 0$ are half the solutions to the Bogoliubov-de Gennes (BdG) Hamiltonian, and where the rest of the BdG spectrum $E_n \leq 0$ are related by particle-hole symmetry. The solutions are

$$\begin{aligned} \phi^{\uparrow,m}_{x,y} &= +U^m_{x,y} \langle \hat{n}^{\downarrow,m}_{x,y} \rangle, \\ \phi^{\downarrow,m}_{x,y} &= +U^m_{x,y} \langle \hat{n}^{\uparrow,m}_{x,y} \rangle, \\ \Phi^m_{x,y} &= -U^m_{x,y} \langle \hat{c}^{\uparrow,m\dagger}_{x,y} \hat{c}^{\downarrow,m}_{x,y} \rangle, \\ \Delta^m_{x,y} &= +U^m_{x,y} \langle \hat{c}^{\uparrow,m}_{x,y} \hat{c}^{\downarrow,m}_{x,y} \rangle, \\ \Delta^{m*}_{x,y} &= +U^m_{x,y} \langle \hat{c}^{\downarrow,m\dagger}_{x,y} \hat{c}^{\uparrow,m\dagger}_{x,y} \rangle. \end{aligned} \tag{5}$$

Note the sign of the Fock terms due Fermionic anticommutation. From these equations, its clear that we expect the mean-fields to go to zero on the Zn sites.

6 Free energy

At zero temperature, the free energy of the interacting theory is

$$\mathcal{F} = \mathcal{F}_{\rm mf} + \langle \mathcal{V} \rangle - \langle \mathcal{V}_{\rm mf} \rangle, \tag{6}$$

where the mean-field free energy is

$$\mathcal{F}_{\rm mf} = E_g - 2k_B T \sum_n \left[\log 1 + e^{-E_n/k_B T} \right], \tag{7}$$

where T is the temperature of the thermal bath enclosing the system, k_B is the Boltzmann constant, E_n are the eigenvalues of the mean-field Bogoloiubov-de Gennes Hamiltonian $\mathcal{H}_{\rm mf}$ and the ground state energy is given by

$$E_g = \sum_n [E_n] + \frac{1}{\mathcal{N}} \sum_\sigma \sum_m \sum_{x,y} \left[\langle \hat{n}^{\sigma,m}_{x,y} \rangle \right], \tag{8}$$

and $\mathcal{N}$ is the number of degrees of freedom. The quartic interaction $\langle \mathcal{V} \rangle$ in the mean-field basis $\{E_n, |n\rangle\}$ is given by

$$\langle \mathcal{V} \rangle = \frac{1}{\mathcal{N}} \sum_\sigma \sum_m \sum_{x,y} U^m_{x,y} \Big[\langle \hat{n}^{\downarrow,m}_{x,y} \rangle \langle \hat{n}^{\uparrow,m}_{x,y} \rangle - \langle \hat{c}^{\uparrow,m\dagger}_{x,y} \hat{c}^{\downarrow,m}_{x,y} \rangle \langle \hat{c}^{\uparrow,m\dagger}_{x,y} \hat{c}^{\downarrow,m}_{x,y} \rangle + \langle \hat{c}^{\uparrow,m}_{x,y} \hat{c}^{\downarrow,m}_{x,y} \rangle \langle \hat{c}^{\downarrow,m\dagger}_{x,y} \hat{c}^{\uparrow,m\dagger}_{x,y} \rangle \Big], \tag{9}$$

The quadratic interaction $\langle \mathcal{V}_{\rm mf} \rangle$ is given by

$$\langle \mathcal{V}_{\rm mf} \rangle = \frac{1}{\mathcal{N}} \sum_\sigma \sum_m \sum_{x,y} \Big[\phi^{\uparrow,m}_{x,y} \langle \hat{n}^{\downarrow,m}_{x,y} \rangle + \phi^{\downarrow,m}_{x,y} \langle \hat{n}^{\uparrow,m}_{x,y} \rangle - \Phi^m_{x,y} \langle \hat{c}^{\uparrow,m\dagger}_{x,y} \hat{c}^{\downarrow,m}_{x,y} \rangle - \Phi^{m*}_{x,y} \langle \hat{c}^{\uparrow,m\dagger}_{x,y} \hat{c}^{\downarrow,m}_{x,y} \rangle + \Delta^m_{x,y} \langle \hat{c}^{\uparrow,m}_{x,y} \hat{c}^{\downarrow,m}_{x,y} \rangle + \Delta^{m*}_{x,y} \langle \hat{c}^{\downarrow,m\dagger}_{x,y} \hat{c}^{\uparrow,m\dagger}_{x,y} \rangle \Big], \tag{10}$$

More details of the inhomogeous self-consistent superconducting density functional theory can be found in Leggett [REF].

7 Green's function

The solution to the Hamiltonian is given by the Green's function, which we remind the reader, is given by

$$\hat{\mathcal{G}}(\omega - i\epsilon, x, y) \equiv \sum_n \left[\frac{\psi_n(x,y)\psi_n(x,y)^\dagger}{\omega - i\epsilon - E_n} \right], \tag{11}$$

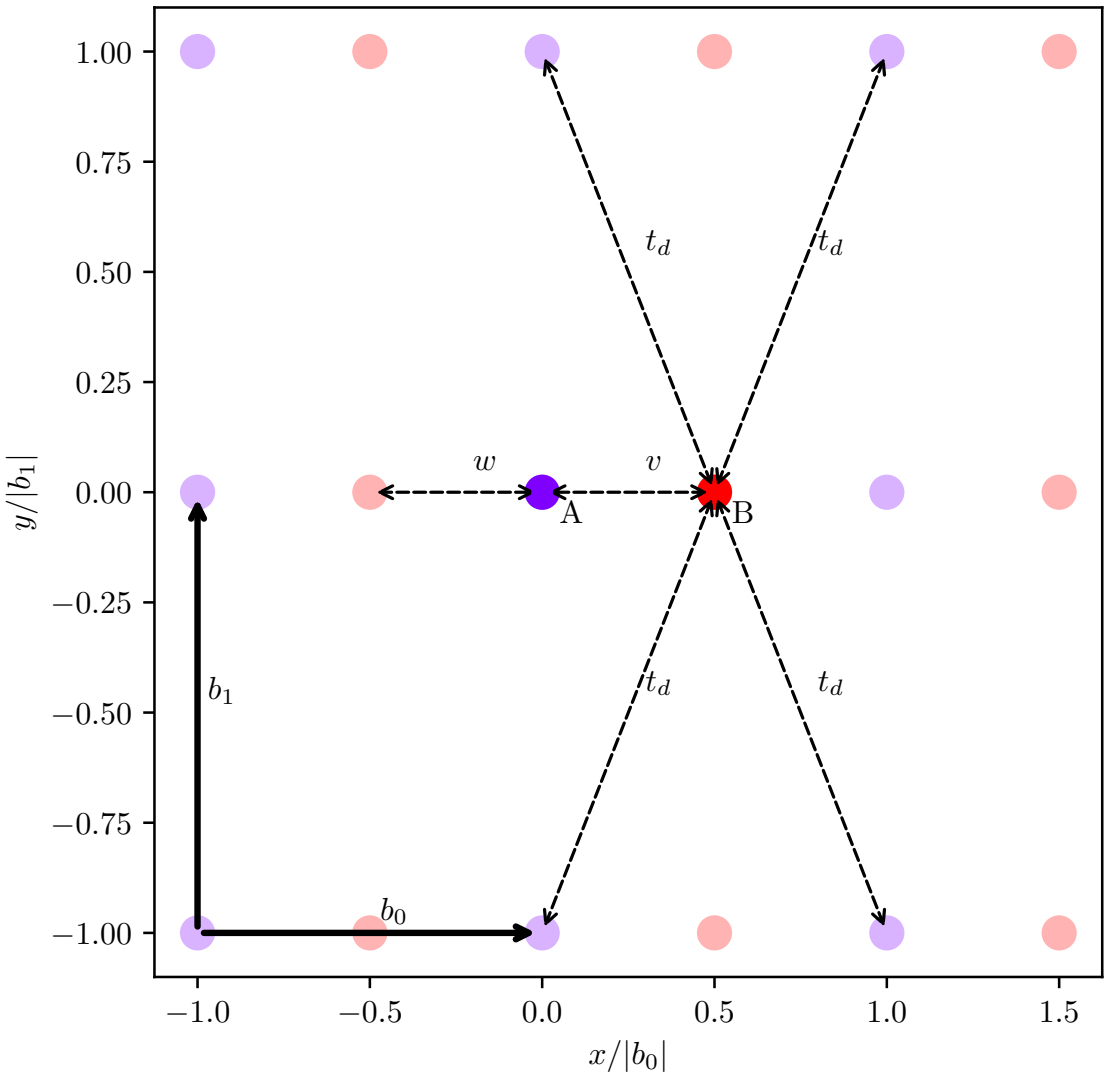

Figure 1: Two-dimensional multiorbital Weyl SSH model. Along the $\hat{x}$-direction, atomic sites A and B within the unit cell form an SSH chain, with intracell hopping v and intercell hopping w. An additional interchain hopping parameter couples the chains into a two-dimensional stack. Following Rosenberg and Manousakis, we study a diagonal hopping t_d, which is a Type II extension of the SSH model in the classification according to Bo-Hung [CITE: Two-Dimensional Extended Su-Schrieffer-Heeger Model; Masters Thesis; Chen, Bo-Hung; July 2018]. The basis for the lattice is given by the vectors b_0 and b_1.

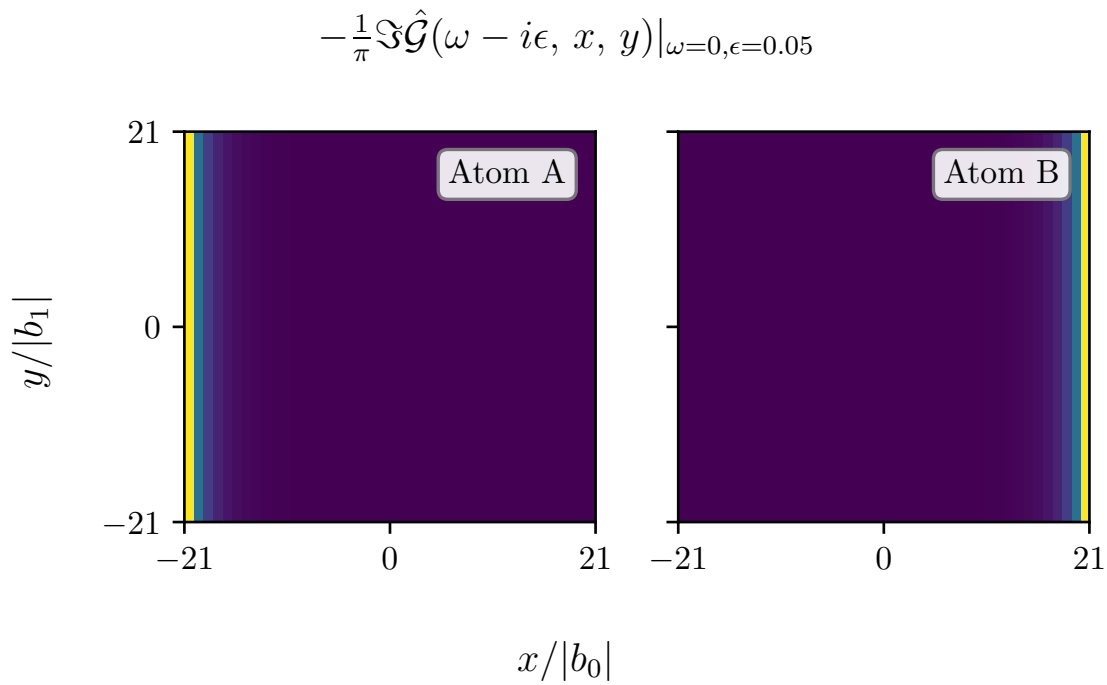

Figure 2: The local density of states of a Weyl SSH model, that is, a spinless square lattice with 43×43 lattice cells, with chemical potential $\mu/t = 0.00$, zero shift $s = 0$ and applied bias $\omega = 0$ and spectral resolution $\epsilon = 0.05$. The SSH chains are in their topological phase $w = 1.2 > v = 0.6$, with diagonal hopping $t_d = 0.9$, which does gives rise to no topological modes perpendicular to the chains. Notice the zero bias density at the left edge for atom A and on the right for atom B, a signature of topological modes.

where $\psi_n(x, y)$ is the eigenvector associated with the nth eigenenergy, resolved at the point on the lattice (x, y). The Green's function matrix has a component for each atom. The Green's function is experimental observables through the *local density of states*,

$$\text{LDOS}(\omega; x, y) \equiv -\frac{1}{\pi}\Im\hat{\mathcal{G}}(\omega - i\epsilon, x, y), \tag{12}$$

which is a measure of the density of states at a given point, at a specified applied bias energy ω and resolution ϵ. The local density of states for each atomic site with the unit cells are given in Figure 2, showing the existence of zero energy edge states.

Resolving the local density states at the level of the unit cell (summing over the atoms) and integrating over the y-axis, we study the *spectrum* of the system. The spectrum shows edge modes present at zero energy at the edges, which are not present in the bulk. These

are distinct signatures of Majorana modes. In addition to localised density at the edge, there exists a 'tail' of density which the authors investigate further.